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Process Analytical Chemistry: Applications of Ultraviolet/Visible Spectrometry in Environmental Analysis: An Overview

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Abstract: Process analyzers based on the selective absorption of the ultraviolet (UV) and/or visible (VIS) radiation provide an on-line monitoring highly reliable of the composition and the industrial process control.

This review provides information about the technological advances in this area and its industrial applications, especially environmental applications.

Keywords: UV/VIS spectroscopy, process analytical chemistry, environmental applications, review

INTRODUCTION

Process analytical chemistry is the application of analytical science for measuring and controlling industrial chemical processes. It began to be applied in the petroleum industries, where the process control was initially based on measures of temperature, pressure, and flow rate. Subsequently, the production of new materials, environmental regulations, etc., demand to have real-time data to ensure the highest product quality, lower costs, and safety for workers, communities, and the environment. Absorption spectroscopy is frequently used with refining industries, petrochemicals, fine chemicals, materials, and pharmaceuticals.

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Because of the importance gained by process analytical chemistry in recent years, optical techniques are very useful because of the ability to assess the processes for control in real time because it is a fundamental requirement to maintain optimum conditions of operation to move in terms of quality, quantity, high level of expertise, cutting spending, and improving efficiencies.

The on-line analysis of the composition of a sample based on selective absorption UV/VIS has been used for monitoring and controlling processes for over 60 years.

The criteria for the design of process analyzers UV/VIS are somewhat different to those of laboratory instruments because analyzers often must work in adverse environmental conditions, such as vibration, wide temperature changes, corrosive atmospheres, etc.

At present, new and improved sources of UV and VIS light, detectors, interference filters, and optical materials allow the expansion of applications of analyzers UV/VIS, and in this regard, analyzer processes that use a photodiodes detector and holographic gratings can be used for multicomponent analysis, which previously was not feasible. In general, the absorption measures of UV/VIS radiation found a great application in the quantitative determination of a wide variety of species, both inorganic and organic.

A Brief History

During the past half-century of design evolution, photometric analyzers based on UV/VIS absorption have found increasingly broad applicability in process analytical chemistry. The UV/VIS analyzers are primarily used in the stream process for direct supervision of the composition and process control closed-circuit.

Before 1930, in Germany, Muller developed one of the first UV analyzers to monitor elemental mercury in air, using the atomic resonance line of mercury at 253.7 nm wavelength measurement (1).

In 1939, Woodson also described an analyzer sensitive to mercury by using a low-pressure discharge lamp using as source for the line 253.7 nm (2).

Until the late 1950s, most analyzers were developed within the petroleum and chemical industry for their own use and some were designed for research facilities at the university. Many of these early instruments used line resonance mercury 253.7 nm obtained from a germicide lamp or were adapted to a spectrophotometer laboratory.

Some of the first UV/VIS analyzers manufactured appropriately, were well used and covered existing needs at the time. However, others were poorly designed for implementation on-line and were often poorly applied. As a result of some of these early failures, passed a generation for some companies accepted the use of analyzers in general and in particular UV/VIS. The photometric analyzer of split beam was developed by the DuPont Company in the

mid-1950s initially to meet the need for accurate monitoring and control of the concentration of chlorine gas in the production process titanium dioxide. In 1962, about 200 of these analyzers were operated within the DuPont Company when the commercial version was introduced as the first product of the newly formed Instrument Products Division. Other UV/VIS photometric analyzers, particularly the photometric analyzer Analytic Systems (later Teledyne) and the UV analyzer Halikainen were also available commercially and were used by various companies at that time. However, is the introduction of beam cut analyzer of DuPont with its numerous applications which expands the use of photometric analyzers in industrial processes.

Type of Process UV/VIS Analyzers

Several types of process analyzers based on the UV/VIS absorption are commercially available, from very simple to highly sophisticated designs.

They can be divided into two basic types:

1. Filter photometric analyzers
 - Single beam
 - Dual wavelength, split beam
 - Filter-wheel chopper
 - Dual-beam
2. Process spectrophotometers
 - Process scanning spectrophotometers
 - Process photodiode array spectrophotometers

Filter Photometric Analyzers

Filter photometric analyzers are the most common class of UV/VIS analyzers and the most basic is the simple beam (Figure 1).

The second type is based on a differential measurement at two wavelengths simultaneously (Figure 2).

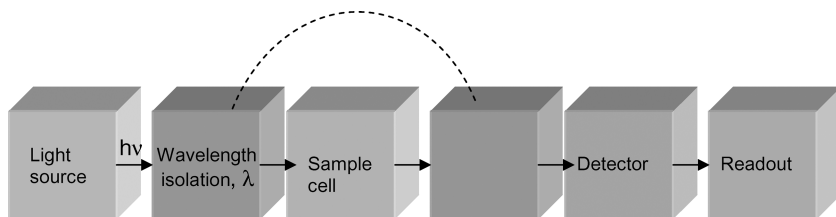


Figure 1. Block diagram of an absorption analyzer.

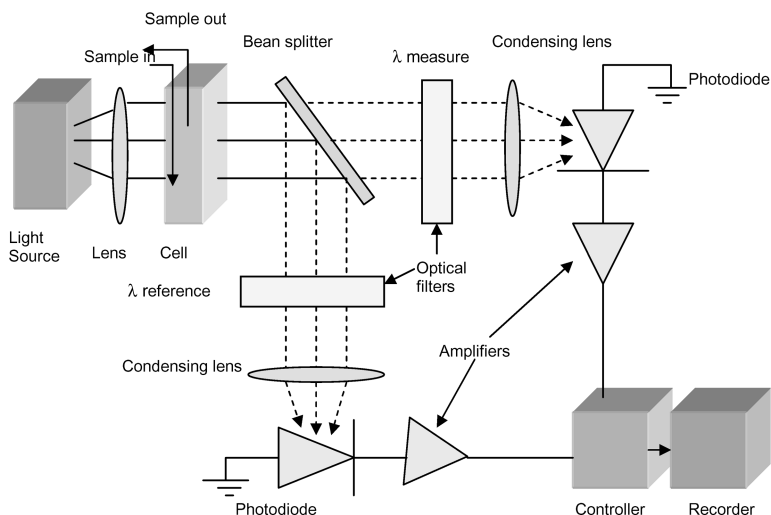


Figure 2. Block diagram of two-wavelength UV-Vis analyzer.

The third type of photometric analyzer is a filter wheel containing two or more interference filters (selected for different wavelengths). The chopper motor rotates the filter-wheel reference and analytical filters alternately and continuously into and out of the optical path. The light source is collimated by a series of lenses and directed through the filters and the cell sample. Through a lens, the light is transmitted to the detector, which provides the signal to each wavelength. It uses two wavelengths to measure a single component. However, if more filters at selected wavelengths are used, multiple concentrations may be used (Figure 3). Other analyzers use up to three separate hollow cathode lamps with a system of mirrors to provide several lines. Lastly, they also employ dual-beam systems.

Process Spectrophotometers

A diagram of a process scanning spectrophotometer UV/VIS is shown in Figure 4. A scanning spectrophotometer cannot be installed in a processing plant with an acceptable degree of reliability, because it has a very delicate optics, but this problem can be solved with the use of optical fibers to conduct transmitted radiation from the sample to the detector. Overall, the monochromator and the detector are installed in a “clean” area with the cell sample at the plant.

An outline of a process diode array analyzer is depicted in Figure 5. This design can be adapted to the use of fiber optics for conducting radiation between the various components of the system (3, 4) (Figure 6).

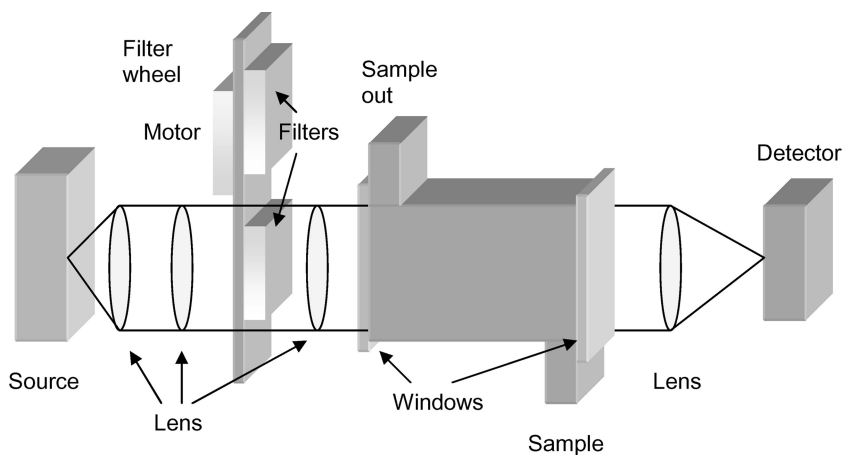


Figure 3. Block diagram of dual-wavelength analyzer with chopper.

Applications

Halogens and Halogenated Compounds

Halogens, particularly chlorine, both in gaseous and liquid, are monitored with UV/VIS analyzers.

Oxidizing Agents

In addition to halogens, other compounds oxidants such as sodium hypochlorite, chlorine dioxide, hydrogen peroxide, ozone, and potassium permanganate are

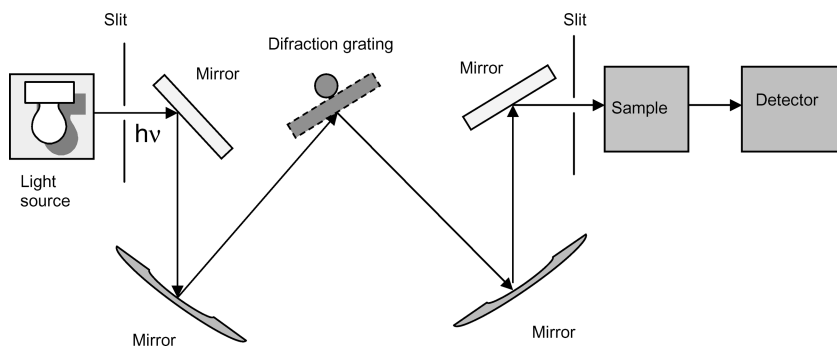


Figure 4. Block diagram of a scanning spectrophotometric analyzer.

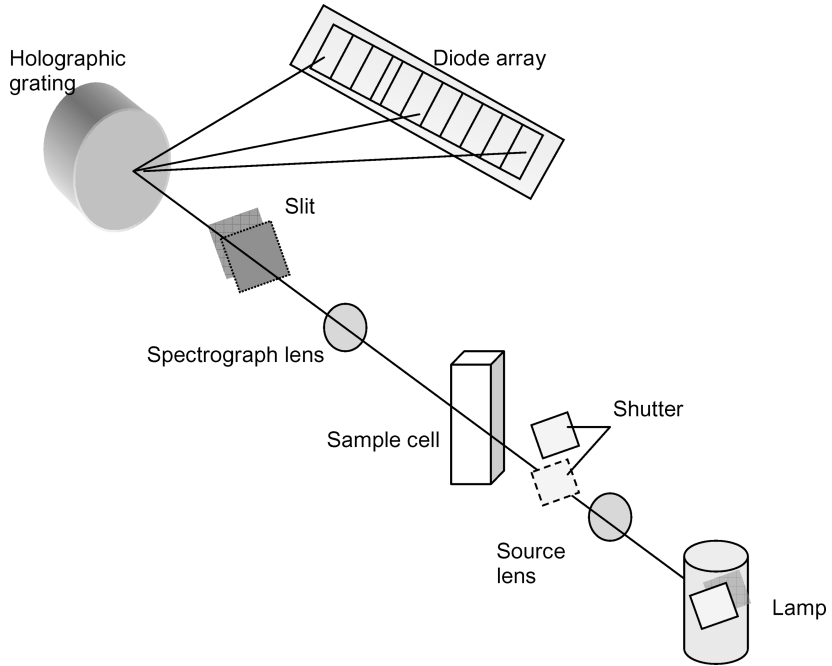


Figure 5. Block diagram of a diode analyzer.

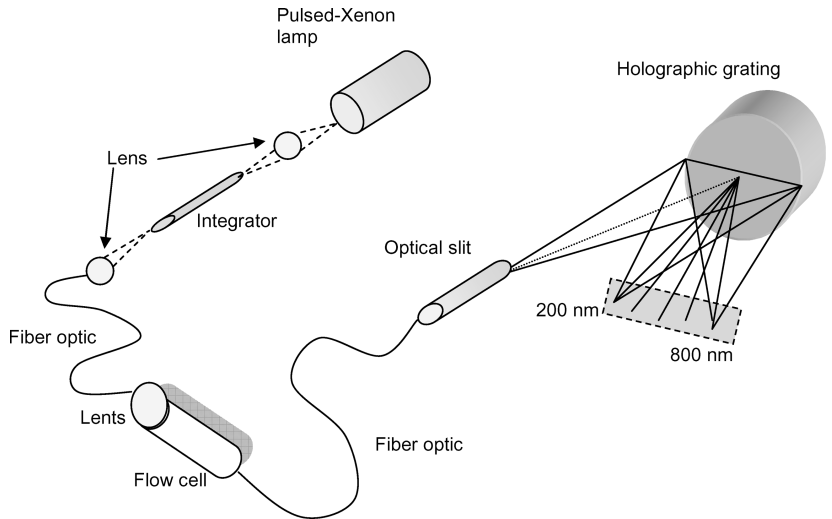


Figure 6. Block diagram of a process diode array analyzer with fiber optic.

examples of compounds whose concentrations can be monitored with UV/VIS analyzers.

Aromatic Compounds

All aromatic compounds are strong absorbers of UV radiation and therefore subject to be analyzed using this region of the spectrum. These analyzers are applicable for the control of organic compounds used in various industries such as textiles.

Color Measurement

Color is a primary quality specification monitored with a visible analyzer. Coloured industrial wastewater may require colour removal before discharge in the receiving medium. In compliance with regulation, wastewater colour measurement or colour difference determination is needed, in reference to an acceptability level. In this sense, Coque et al. (5) developed a spectrophotometric method for monitoring processes bleaching different colors of visible spectrum. The method uses photolysis and photooxidation as testing methods for the disposal of color. These experiments were carried out in a photochemical reactor with a volume of 7.5 mL formed by a low-pressure mercury lamp surrounded by a coil of quartz. This process, which is simple and fast, allows control of the separation processes of color and requires only the acquisition of an on-line visible spectrum in front treatment time.

Kandelbauer et al. (6) accomplished a study of bleaching dye with an enzyme immobilized in a reactor using on-line spectroscopy. In this model, the enzyme reactor was equipped with several spectroscopic UV/VIS sensors allowing on-line control of reactions bleaching.

Valadez-Blanco et al. (7) carried out a study of the color control in-line during the food extrusion. The use of a spectrophotometer equipped with fiber optics for color monitoring is useful for the characterization of products and effective control over them.

Sulphur Compounds

Sulphur-containing compounds often strongly absorb UV radiation. H_2S , COS, CS_2 , and SO_2 at low concentration are monitored and controlled for safety, environmental compliance, and process efficiency. The measurement of SO_2 is probably the largest application of UV analyzers. Applications for monitoring low concentrations of H_2S in hydrocarbon streams have been successful, but at times background absorbance, often from trace aromatics, interferes with the measurement. With process spectrophotometers, for example, process photodiode array analyzers, compensation for interferences can be provided by a chemometrics software program in the analyzer computer.

The strong absorption of UV radiation from H_2S and SO_2 was used in an analyzer system developed to monitor the relationship of their concentrations in a recovery sulphur plant. A ratio of 2:1 concentrations of H_2S to SO_2 should be monitored for maximum sulphur production and the minimum SO_2 emission in the atmosphere. Two photometric analyzers are used in this system, one for each type of gas. The potential interferences and calibration techniques are discussed in this work, one of the first collected in the literature (8).

Lang et al. (9) realized a study to monitor the volatile sulphur compounds in factories producing sulphite. These researchers concluded that the major sources of air pollution could be influenced significantly by controlling operating conditions. Maintaining optimal operations depends on reliable data to the operators provided at a high degree of continuity. A system of SO_2 analyzer and completely reduced sulphur compounds was tested extensively in the operations of many factories.

Saltzman (10) developed an important study on a continuous photometric analysis to measure and control the sulphur recovery operations. This system is being used at present to control and monitor the relationship of SO_2 and H_2S in the tail gas of sulphur recovery plants. The rapid and continuous answers of these analysis systems have been highly effective in improving efficiency and reducing sulphur emissions into the air.

Saltzman and Dell (11) developed photometric analysis systems for controlling low levels of H_2S . Also, Saltzman and Bilinski (12) presented an analysis of the tail gas and control to the recovery of sulphur. Finally, Saltzman (13) developed a system to monitor low levels of H_2S in streams with alkanes. This has resulted in the emergence of a system for on-line analysis based on the measurement of intense UV absorption of chains amine with H_2S .

Srinivasan and Grutzeck (14) carried out a study of the adsorption of SO_2 by zeolites that were synthesized from fly ash. The coal combustion processes that produce a large number of rich sulphur compounds are very harmful to the atmosphere. In the past, SO_2 emissions were not regulated and as a result there was an increase of this species in the atmosphere and an increase in acid rain. For this reason, SO_2 concentrations in the atmosphere are now regulated by the states. In this system a gas containing 2000 ppm of SO_2 , 10% O_2 , 10% CO_2 , and 80% N_2 was used. The device consisted of a cell adsorption designed to contain samples of zeolite and a spectrophotometer to record concentrations of SO_2 in the gas stream. The cell adsorption was 10 cm long and 1 cm in diameter.

Wallace et al. (15) have recently attempted to obtain a colorimetric response to H_2S . It is known that H_2S is a poisonous gas that is generated in many industrial processes, generally from petroleum-related industries. There has been a development of thin films containing various species of metal to detect low concentrations of H_2S . The most commonly used films are made of copper, silver, tin dioxide, and cadmium sulphur. These films were fabricated with a 3.0% (w/w) concentration of copper chromophores, by immobilizing in an

acrylate-based polymer. In the exposure, H_2S gas diffuses into the film and reacts with copper causing a change of color in the film.

Inorganic Compounds

Salts of transition metals such as iron, copper, chromium, manganese, and nickel are often strong absorbers of UV and/or VIS light. Typical applications include measurement and control of ferrous ions concentrations in steel pickling baths, cupric ions in nitric acid production, Cr(III) and Cr(VI) in aluminum plating processes, and a number of metallic ions in plating baths.

A photometric analyzer was developed to selectively monitor concentrations of Cr(VI) in the range of 0–100 ppb in wastewaters. Background absorption is measured using as reference a sample where Cr(VI) has been reduced to Cr(III) with SO_2 (16).

Pharmaceuticals

Many pharmaceutical compounds strongly absorb UV radiation. During production, several process steps can be controlled by measurement of the concentrations with UV analyzers. For example, a UV analyzer can be used to monitor the procedure for removing impurities (cleaning step). This analyzer is also used in other industries such as food and especially in the chemical industries.

Fermentation Processes Monitoring

Noui et al. (17) developed several high-resolution spectrophotometers suitable for at-line control of fermentation processes. High-resolution spectrophotometers suitable for use in at-line monitoring of upstream or downstream fermentation processes are becoming more important in process control as a way of improving product quality and production efficiency. During the past few years many attempts have been made to adapt conventional analytical laboratory instrumentation for process control applications. However, these techniques are either too complex to be applied to industrial processes or are restricted to one measurement channel only. With recent advances in electronics and particularly in light detectors, a new generation of spectrometers has been developed.

Environmental Control

Analysis of Gaseous Pollutants in Air: The main air pollutants are ozone, which we will talk about later, nitrogen oxides, sulphur dioxide, carbon monoxide, lead, particulate matter, and organic pollutants. Some of these can be measured and monitored using UV/VIS spectrophotometry.

Nitrogen oxides (NO_x) are a mixture of nitric oxide (NO) and nitrogen dioxide (NO₂) and are produced from natural sources, motor vehicles, and other combustion processes of fuels.

Armending et al. (18) carried out a study to take local control of NO₂ concentrations in ambient air by absorption spectroscopy. This method was trying to control the tropospheric NO₂ using the differential absorption at 600 nm. For in situ measurement of NO₂ use an instrument called MOAS (multipass optical absorption spectroscopy) with high spatial and temporal resolution and sensitivity.

Subsequently, Armending et al. developed a quick scan with a laser spectrometer to determine atmospheric trace compounds and in particular traces of OH⁻ (19). This type of spectrometer is used to measure simultaneously OH⁻, SO₂, CH₂O, and naphthalene in the UV and NO₂ in the visible.

Using the scanning imaging absorption spectrometer for atmospheric cartography (SCIAMACHY) pre-flight model satellite spectrometer, gas-phase absorption spectra of the most important atmospheric trace gases (O₃, NO₂, SO₂, O₂, OClO, H₂CO, H₂O, CO, CO₂, CH₄, and N₂O) have been measured in the 230–2380 nm range at medium spectral resolution and at several temperatures between 203 and 293 K by Bogumil et al. (20). The observation of the changing atmosphere requires powerful observational techniques that enable global monitoring of atmospheric trace gas concentrations with high spatial resolution.

A new generation of optical remote-sensing instruments for monitoring of atmospheric trace gases from space has been developed in the last decade (1990–2000). These satellite instruments operate in the ultraviolet, visible, and near-infrared regions of the spectrum and use backscattered and surface- or cloud-reflected sunlight to detect several important atmospheric constituents such as O₃, NO₂, SO₂, H₂CO, OClO, BrO, and H₂O using their characteristic absorption features in the UV-Vis region.

Differential optical absorption spectrometry (DOAS) is a remote-sensing technique. The analytical principle of the DOAS system is based on the light absorption in the near-UV and Vis regions of those pollutants with fine vibrational structures such as SO₂, NO₂, O₃, nitrous acid, formaldehyde, benzene, toluene, etc. The relationship among the adsorbed light intensity and the compound concentrations is ruled by Lambert-Beer's law. A UV/Vis spectrometer was installed at Mt. Cimone Station in 1993. Since then it has carried out zenith scattered solar radiation measurements at sunrise and sunset in the 407–464 nm spectral region. Data have been processed through DOAS methodology in order to obtain NO₂ slant column by Petritoli et al. (21). A long path differential optical absorption spectrometer (LP-DOAS) system has been developed to measure simultaneously atmospheric trace gases concentration and spectral extinction coefficient in the visible region over a few hundred meters by Lee and Kim (22). Concentrations of atmospheric gases, SO₂, NO₂, and O₃ are evaluated from the differential optical absorption spectra measured by system.

Performance of the system was tested in the field over a 1500-m path length. Avino et al. (23) have reported and discussed the behaviors of NO_2 , O_3 , HNO_2 , formaldehyde, benzene, and toluene measured by this system during the 2000–2003 period in the Rome area (an area highly influenced by anthropogenic sources) and in Pietracupa area (considered a remote site). The results show how the DOAS system highlights very well the secondary pollution evolution.

Dooly et al. (24, 25) conducted a study on the control of exhaust gases using UV based on the same system. It described a fiber optic sensor for controlling hazardous exhaust gases, such as NO_2 , NO, and SO_2 .

Evangelisti et al. (26) measured concentrations of SO_2 , NO_2 , O_3 , and HNO_2 in the atmosphere using a system called GASCOAD (gas analyzer spectrometer correlating optical absorption differences).

Levelt et al. (27–29) and Levelt et al. (28, 29) present the ozone monitoring instrument (OMI). OMI is a UV/VIS nadir solar backscatter spectrometer, which provides nearly global coverage in one day with a spatial resolution of $13 \text{ km} \times 24 \text{ km}$. Trace gases measured include O_3 , NO_2 , SO_2 , HCHO, BrO, and OCIO. In addition, OMI will measure aerosol characteristics, cloud top heights, and UV irradiance at the surface.

Giovanelli et al. (30) present three methods for the validation of the SCIAMACHY using the UV/Vis. A DOAS spectrometer, called GASCOD/A4, installed onboard the M55 aircraft, which can fly up to 20 km of altitude, was used. The first method is the classic one of the criss-crossing that requires the temporal and spatial simultaneity of the measurements onboard both the ENVISAT-1 satellite and the M55 aircraft. The second one is based on a Lagrangian approach, which allows for the identification of the best flight trajectory of the M55, on the basis of the dynamic and chemical “history” of the air masses at the aircraft altitude. In this case the comparison can be extended for longer time periods. The latter method is proposed to validate the tropospheric amounts of O_3 and NO_2 .

Water Analysis. During the 1980s, the introduction of miniature diode array detectors, combined with powerful microprocessors and state-of-the-art mathematical tools, led to a renaissance of UV-Vis spectrometry and enabled the rapid spread of compact, relatively low-cost yet still powerful laboratory UV-Vis machines. During the 1990s, the technology moved from the laboratory into the field, but still in the form of relatively large, complicated and expensive off-line analysers.

A submersible UV/VIS spectrometer for in situ real-time measurements is presented by Langergraber et al. (31). The UV/VIS spectrometer proved to be a promising instrument to quantify wastewater pollutant loads for integrated management and control of municipal sewer networks and monitoring of the treatment plant itself. The submersible spectrometer utilizes the UV/VIS range (200–750 nm) for simultaneous measurement of organic matter, suspended solids, and nitrate with just a single instrument. The miniaturized UV/VIS

spectrometer is a probe installed directly in the process. Therefore, it requires no sampling, no sample preparation, and no reagents. The spectrometer is equipped with an auto-cleaning system. A global calibration for typical municipal wastewater is provided as default configuration of the UV/VIS spectrometer. The submersible UV/VIS spectrometer is a spectrometric probe of 44 mm diameter and about 0.6 m length. The measurement takes place directly in situ without sampling or sample treatment. A single measurement typically takes about 15 s. The instrument is equipped with an autocleaning system using pressurized air. Due to the compact size the sensor can be easily applied in 2" bore holes; for example, for groundwater monitoring. An ex-proofed version is available for sewer applications. The instrument is a 2-beam 256 pixel UV/VIS spectrometer, with a xenon lamp as a light source. Onboard electronics control the entire measurement procedure. All of the controller electronics are included in the 44-mm tubular anodized aluminum housing and include a data logger and a water level meter. The low power consumption eases field application by means of battery or solar power supply. The probe has a data logger on board, capable of storing, e.g., complete absorption spectra for 1 month at a measuring interval of 30 min.

For many years, UV/VIS spectroscopy has been proposed as an alternative and a quick way to estimate the level of pollution from sewage and surface water. The wavelength range between 200 and 300 nm has a particular interest. 254 nm was selected due to the correlation existing between the absorbance and total organic carbon (TOC) content (32), between UV absorbance and organic matter content (33), and between absorbance and chemical oxygen demand (COD) (34).

Multi-Parametric Measurements by In Situ and On-Line UV/VIS Spectroscopy

During the 1980s, the introduction of miniature diode array detectors, combined with powerful microprocessors and state-of-the-art mathematical tools, led to a renaissance of UV/VIS spectrometry that enabled the rapid spread of compact, relatively low-cost, yet still powerful, laboratory UV/VIS machines. During the 1990s, the technology moved from the laboratory into the field but still in the form of relatively large, complicated, and expensive off-line analyzers. However, the market for in situ optical sensors/probes was still, and still is, dominated by relatively simple photometers, which are able to measure only one or two wavelengths at a time. These instruments are, thus, limited to measuring one parameter only and generally employ quite crude and unstable methods to compensate for cross-sensitivities to variations; for example, in a water matrix. A lot of additional information, however, as well as reduced cross-sensitivity, can be obtained when using the entire absorption spectrum instead of single wavelengths. The development in electronics and optics over recent years has enabled the marriage between full-spectrum UV/VIS spectroscopy and small-scale instruments. Moreover, these developments have allowed the design

of robust spectroscopic analyzers that can function in harsh environments, such as inside sewer and industrial process streams, with little or no maintenance because neither chemicals nor moving parts for cleaning are necessary for their functioning.

The technique has been used to estimate the rapid in situ parameters of inorganic and organic urban wastewater. In this regard, the research group of Matsché (35, 36) investigated the application of UV absorption at $\lambda = 254$ nm in many treatment plants for the measurement of organic pollution in the influent and in the effluent. The UV probe used measures the absorption at $\lambda = 254$ nm, which is typical for aromatic compounds and has a correction for turbidity and color at $\lambda = 350$ nm. The combination of these measurements leads to reliable results, which have a good correlation to both COD and TOC for domestic wastewater. This works very well on wastewater treatment plants, but in sewer systems there are some special problems.

The phenomenon of wastewater sample aging is supposed to be treated with a low-temperature auto-sampler. Baurès et al. (37, 38) present two examples of treated wastewater, the quality of which varies with the time of conservation. This evolution may pose some problems with respect to regulation compliance or process control. After the explanation of the mechanisms involved in sample aging, some recommendations are proposed in order to improve the UV off-line measurement of TSS and COD of treated wastewater.

Wastewater Fingerprinting

Municipal wastewater is a mixture of sewage from residential and office buildings, restaurants, schools, hospitals, factories, etc. Its variations with time reflect the human activity as modified by weather conditions. For many years UV/VIS spectroscopy has been proposed as an alternative and rapid way to estimate the level of pollution of wastewater and surface water. In this sense, municipal wastewater samples have been collected by Wu et al. (39) in three different types of communities and fingerprinted by optical methods combining UV/VIS spectrometry, synchronous fluorescence spectrometry, and turbidity.

The spectra, referred to as fingerprints, obtained with such on-line spectrometers are used for the characterization of the sampled water. Within these sometimes almost featureless fingerprints, one can find a huge amount of information about the water composition. The fingerprints are used to monitor changes in the water composition through analysis of the general shape of the spectrum or absorption at a specific wavelength. Furthermore, they are used to derive more specific parameters, such as turbidity, nitrate concentration, and sum parameters such as spectral absorption coefficient at 254 nm (SAC254), total organic carbon (TOC), and dissolved organic carbon (DOC), which are commonplace in water analysis (31, 40).

Turbidity, due to suspended substances, causes light scattering and shading, thus influencing the absorption over the entire fingerprint. This is an important factor that influences in situ measurements and requires compensation in order to obtain reliable and reproducible readings. An algorithm for compensation of turbidity has been developed, based on a relationship between scattering intensity and wavelength as a function of the particle diameter and the spectral shape caused by suspended solids (41). The scattering is dependent on wavelength (42).

Monitoring and control of sequencing batch reactors (SBRs) is currently a very active field of research. Langergraber et al. (43) suggest using the submersible UV/VIS spectrometer, described above, for on-line monitoring and control of an SBR. It is installed directly in the reactor and measures several parameters simultaneously in real time. Using nitrate nitrogen measurements it is possible to monitor and control nitrification and denitrification. Suspended solids measurements can be used to optimise settling times. The paper shows calibration results and time series for typical SBR cycles.

Based on novel technologies, a modular online monitoring station suitable for continuous application in sewer networks, wastewater treatment plants, and receiving water bodies has been designed by Gruber et al. (44).

On the other hand, the use of UV/VIS spectroscopy for water quality measurements is based on the solution of the correlation between the surrogate parameter absorbance and the resulting equivalence parameters. The coherence of absorbance and equivalence parameters (COD_{tot} , COD_{sol} , TSS) is solved by Hochedlinger et al. (45) with different regression methods. The correlation of absorbance and concentrations is analyzed based on linear regression methods, model tree regressions, multivariate regression methods, and support vector machines using sequential minimal optimization algorithm.

A submersible UV/VIS spectrometer was used to monitor a paper mill wastewater treatment plant (42). It utilizes the UV/VIS range (200–750 nm) for simultaneous measurement of COD, filtered COD, TSS, and nitrate with just a single instrument. The instrument measures in situ, directly in the process. Paper mill wastewater shows typical and reproducible spectra at various process measuring points. There is a relative maximum at 280 nm due to the absorbance by dissolved organic substances, mainly ligninic acids. Comparison of absorbance spectra distinctly shows the decrease of this peak, indicating biological degradation throughout the treatment process. A global calibration for paper mill wastewater is provided as default configuration of the instrument. Usually high precision can be achieved using this standard parameter set. Calibrations to conventional parameters are now merely carried out for purposes of reference-checking. Only the spectral information can also be used for plant control.

Langergraber et al. also present a method to define alarm parameters from spectral data (46). Alarm parameters are in many ways different from measurements of well-defined chemical substances. Being confronted with an

increasing number of potentially harmful compounds as well as financial and logistic constraints, new variables (such as alarm parameters) that allow for an integrated assessment or for a first screening can be a solution. Spectral data and their evolution over time are rich in information and compensate for losses due to aggregation and generalization.

Different calibration approaches are applied to UV/VIS spectra measured by in situ sensors installed in the influent to the biological stage of six wastewater treatment plants by Rieger et al. (47). Three calibration methods were applied to UV/VIS spectra recorded in the influent of six wastewater treatment plants to measure total COD (COD_{tot}), filtered COD (COD_{fil}), nitrate and nitrite nitrogen (NO_x-N), and total suspended solids (TSS).

The variability of pollution transport in combined sewers is investigated by Hochelinger et al. (48) using in situ UV/VIS spectroscopy and in a bypass by means of ion selective probes. Additionally, inflow and spill-flow are measured continuously. It is difficult to determine accurate and reliable overflow loads due to the variability of concentrations and flows in a combined sewer.

In other work, Langergraber et al. (49) describe how alarm parameters can be derived from spectral data.

A UV/VIS spectrometer has been installed in June 2004 at the outlet of the Ecully catchments in Lyon, France (50). After the construction of in situ equipment and the definition of a calibration protocol on wastewater samples analyzed in the laboratory, the UV/VIS spectrometer has been used to measure simultaneously and continuously the TSS and COD equivalent concentrations (in mg/L). The authors present the equipment used, the installation and its maintenance, the calibration results, and measured pollution graphs.

A procedure for the fast estimation of TOC in marine areas receiving urban sewage is described by Coulomb et al. (51). The TOC determination was achieved by the exploitation of UV response of samples related to well-known reference spectra. The reference spectra and deconvolution parameters are discussed, and the results are compared with TOC reference method measurements.

Recently, Jeong et al. (52) developed a study whose objective was to test the feasibility of a new method to improve the accuracy in the estimation of sewage components. Adding to the regression of sewage components with UV absorbance values, the proposed method considered an unclear but existing relationship among characteristics of sewage production. Sewage production showed very defined profiles due to the daily human activities. So the main idea was the combination of measuring the UV absorbance values and analyzing the characteristics of the sewage production. For this purpose, 446 sewage samples taken at 2-h intervals for 51 days at a wastewater treatment plant were statistically analyzed using a neural network. The proposed method could estimate concentrations of total nitrogen (TN) and total phosphate (TP) within practical accuracies as well as total suspended solids.

The purpose of the work carried out by Vaillant et al. (53) was therefore to develop simple qualitative characterization tools based on UV spectra exploitation. The selected procedures only require spreadsheet software and are meant to emphasize qualitative differences or variations. Simple comparisons of spectra helped understand the UV physical response of total suspended solids and enabled checking of wastewater quality conservation.

As can be seen, rapid methods for the *in situ* evaluation of the organic load have recently been developed and successfully implemented in municipal wastewater treatment systems. Their direct application to winery wastewater treatment is questionable due to substantial differences between municipal and winery wastewater. Recently, Carvallo et al. (54) critically evaluated the use of UV/VIS spectrometry in winery wastewater treatment. Three types of samples were considered for the evaluation: (i) actual winery wastewater, (ii) synthetic winery wastewater, and (iii) samples from biological batch reactors treating synthetic winery wastewater composed of a mixture of glucose/ethanol and glucose/tartaric acid.

Biochemical oxygen demand (BOD) is defined as the potential for removal of oxygen from water, by aerobic heterotrophic bacteria that utilize organic matter for the production of energy and for multiplication. BOD values of farm wastes and effluents are generally high but can vary markedly. BOD is widely used to detect and assess contamination of surface waters by organic wastes. In this way, Brookman (55) investigated the estimation of BOD in slurry and effluents using UV spectrophotometry. The absorbance at 280 nm was determined for samples of dirty water and pig and cow slurry effluents collected annually over a period of 3 years. Regression analysis showed an exponential relationship between absorbance and BOD. Well-fitted regressions of absorbance (280 nm) on BOD were obtained for each year, especially for exponential regression analysis.

In 2006, the Langergraber group published a brief but interesting revision about on-line and *in situ* UV/VIS spectroscopy for multi-parameter measurements (56).

UV/VIS spectroscopy has recently been applied for monitoring soil. An example of this application is proposed by Maleki et al. (57), who use a portable analyzer to measure the phosphorus in the soil.

Portable Instrumentation for Field Applications

Companies manufacture on-line, at-line, and in-line instruments. Some spectrophotometers are modular, useful for many applications in the range UV/VIS/NIR. The measurement systems are extremely robust and provide exceptional flexibility due to the use of fiber optics. These systems are fast, efficient, accurate, and include multiple versions, which makes them useful

for a wide variety of applications. Their applications are numerous (moisture, protein, color, etc.) with the field measurements possibility.

The more important applications are

1. Foods:

- Determination of fat, starch, and proteins
- Detection of the moisture of products
- Monitoring of drying processes
- Incoming inspection of powder-based products such as flour or milk powder

2. Measuring running conveyor belts:

- Determination of moisture and color in paper manufacture
- On-line color of textile and plastic lines
- Determination of color and heat protection degree of architectural glass
- Measurement of layer thickness of foils

3. Optical industry:

- Reflection and transmission properties of coated glass
- Colour properties of optical coatings

4. Plastics technology:

- Identification of plastics
- Color of plastic lines or components
- Specification of layer thickness of transparent coatings

5. Pharmaceutical industry:

- Identification of raw materials
- Powder blending

A relatively new study about the potential of these analyzers for on-site and on-line analysis was performed by Saeys et al. (58).

CONCLUSIONS

The advantages of on-line sensors for process control and quality control of samples and industrial products (raw materials and finished products) are widely recognized. The actual number of uses, however, has been somewhat limited due to the limited capacities of available instruments. At present, the introduction of multi-parametric spectroscopic probes with low maintenance

requirements, however, are changing the scope of oversight in line significantly. Moreover, due to the probes spectrometers record the full spectrum, more advanced features that the classics parameters measure are possible. The development of new variables such as the so-called alarm parameters, allowing real-time detection of changes in water quality, is now possible.

On-line UV/VIS spectrometry has proven to be a highly capable and valuable tool for monitoring water quality in real time and in the field. Using one instrument only, it is possible to measure several parameters on-line and in various kinds of liquids, which is a unique feature among online sensor systems. As technology progresses and more experience is gathered, an even further broadening of applications and an increase of the number of parameters that can be measured using this technique will take place.

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